Tomáš Gonda

Researcher in Applied Category Theory and Quantum Information Theory

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Google Scholar

Education

University of Waterloo, Physics — Waterloo, ON, Canada — Jan 2018 – Dec 2023

- PhD thesis on 'Categorical Quantum Mechanics and its Applications in Quantum Information Theory'.
- Published several papers in top-tier journals and conferences in the field of quantum information theory and applied category theory.

Experience

University of Innsbruck, Senior postdoctoral researcher — Innsbruck, Austria — Mar 2025 – present — 1 year 1 month

Conducting independent research in the field of applied category theory and quantum information theory.

- Algebra group, led by Tim Netzer
- Funded by my FWF ESPRIT Grant Ergodic Theory in Categorical Probability

University of Innsbruck, Postdoctoral researcher — Innsbruck, Austria — Jan 2022 – Feb 2025 — 3 years 2 months

Conducting research in applied category theory and quantum information theory.

- Mathematical quantum physics group, led by Gemma De Les Coves

University of Waterloo, Teaching assistant — Waterloo, ON, Canada — 2017 – 2021 — 4 years

Conducting tutorials for graduate level courses in Quantum Theory and Statistical Physics.

ETH Zürich, Visiting researcher — Zürich, Switzerland — Summer 2019

Collaborating with researchers in the group of Renato Renner on topics in information theory.

InstaEDU, Online tutor — Remote — 2013 – 2016 — 3 years

Online tutoring of undergraduate students in science and engineering.

University of Edinburgh, Career development internship — Edinburgh, Scotland — Summer 2015

A summer project on modelling turbulence in Matlab.

- Topic: Transition to turbulence in plane Couette flow

Personal tutor — Bratislava, Slovakia — 2012 – 2013 — 1 year

In-person tutoring of high school students.

Volunteering

Positive Futures Festival, Organizer — Innsbruck, Austria — 2024 – present

Co-organizing an annual two week long music festival with ~40 concerts and ~1000 attendees.

Touch&Play Cologne Festival, Organizer — Germany — 2025 – present

Co-organizing and facilitating an annual week long retreat with ~100 attendees.